

Representing Functions and Fields

Finite representations and approximation error

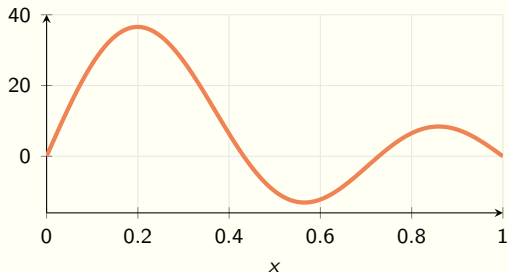
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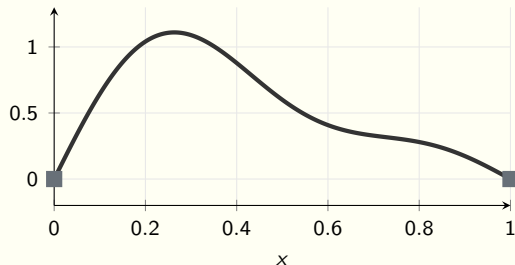
krishnak@utexas.edu

A taut string under a load, $\mathcal{L}u = f$

the load $f(x) = -u''(x)$



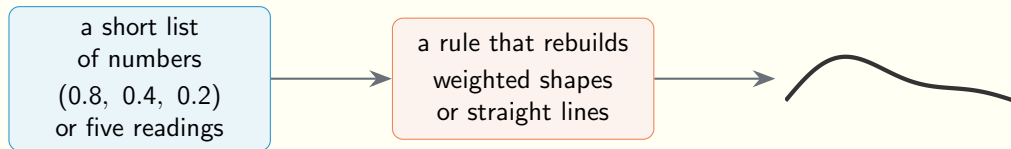
the displacement $u(x)$, pinned at both ends



$$-\frac{d^2 u}{dx^2} = f(x), \quad u(0) = u(1) = 0, \quad \mathcal{L} = -\frac{d^2}{dx^2}$$

$$u(x) = 0.8 \sin(\pi x) + 0.4 \sin(2\pi x) + 0.2 \sin(3\pi x)$$

A finite list of numbers and a rule that rebuilds the function

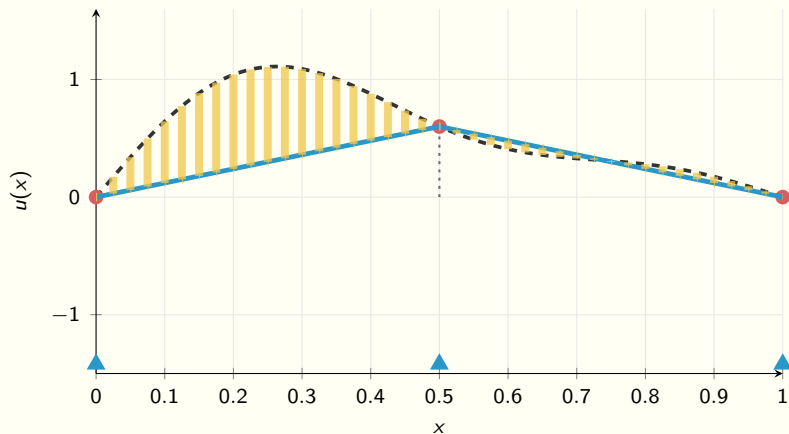


How well can finitely many numbers stand in for a function, and what decides the accuracy?

Outline

- 1 Pointwise sampling
- 2 Functions as vectors
- 3 Norms and the meaning of error
- 4 Inner products, projection, and orthogonality
- 5 Changing the basis
- 6 Convergence, density, and completeness
- 7 Kernels and point evaluation
- 8 Noisy observations, ridge regression, and Gaussian conditioning
- 9 Approximating a family of functions
- 10 Adaptive representations
- 11 Choosing a representation for the cooling fin
- 12 From functions to fields

Sampling the displacement at fixed stations

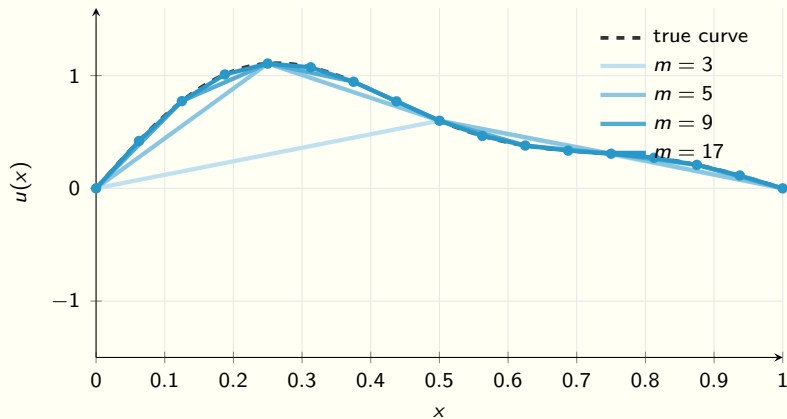


three readings
0, 0.60, 0

worst miss, three sensors
0.82, or **73.5 percent**

[Try it live](#)

Convergence of piecewise-linear interpolation with the spacing



$m = 3, h = 1/2$

73.5 percent

$m = 5, h = 1/4$

**20.1 percent, a
factor of 3.7**

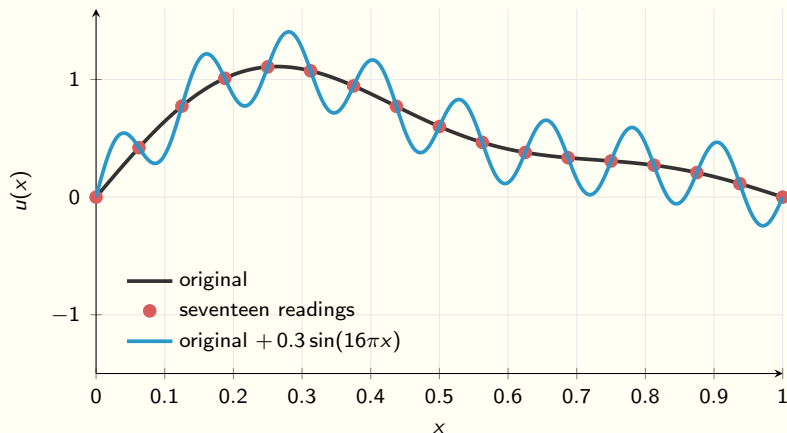
$m = 9, h = 1/8$

**6.3 percent, a factor
of 3.2**

$m = 17, h = 1/16$

**1.6 percent, a factor
of 4.0**

Aliasing of a wave that vanishes at every station



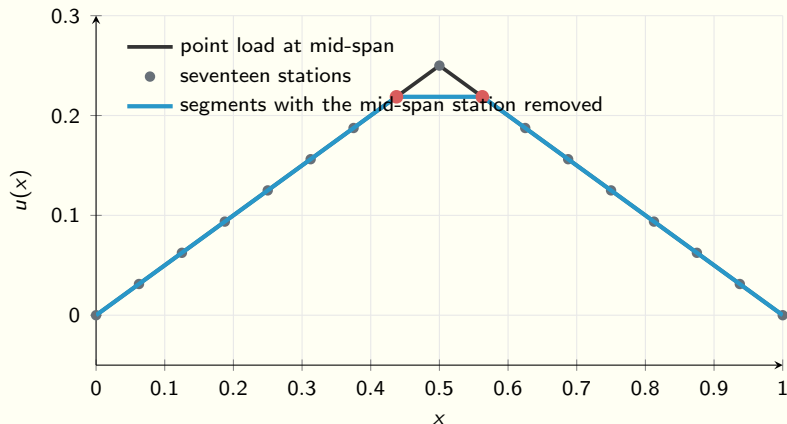
change in the seventeen
readings

0

change at each crest
**0.3, about a quarter
of the largest
displacement**

[Try it live](#)

Grid dependence of sampled representations



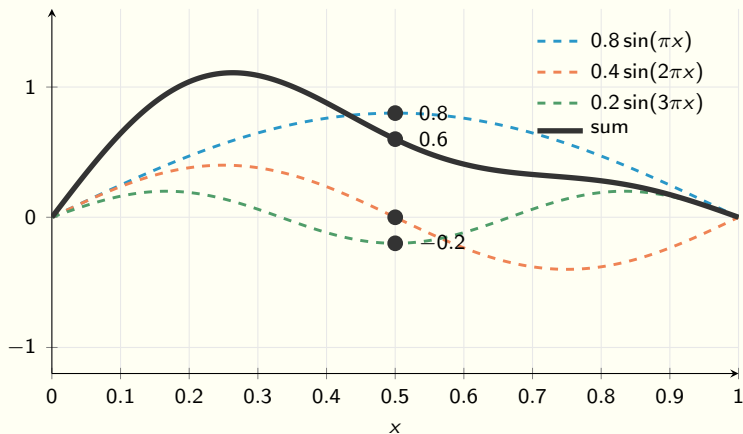
corner, mid-span sensor
removed

**worst miss 0 to 12.5
percent**

smooth displacement,
same sensor removed

1.6 to 1.7 percent

Representing a function with coefficients on sine modes



the whole curve

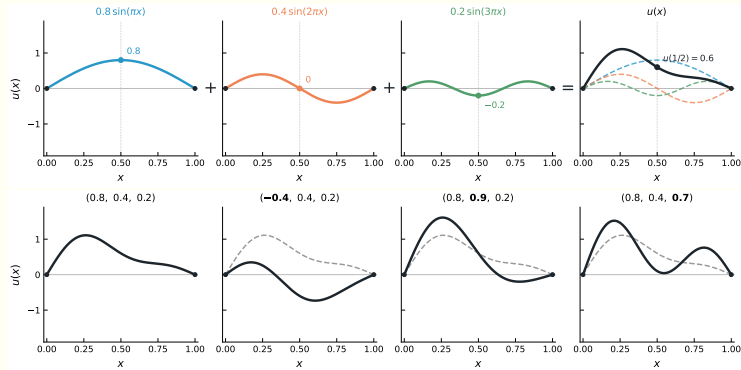
three numbers,
 $(0.8, 0.4, 0.2)$

check at mid-span

$$u\left(\frac{1}{2}\right) = 0.8 - 0.2 = 0.6$$

[Try it live](#)

A three-parameter family of string displacements



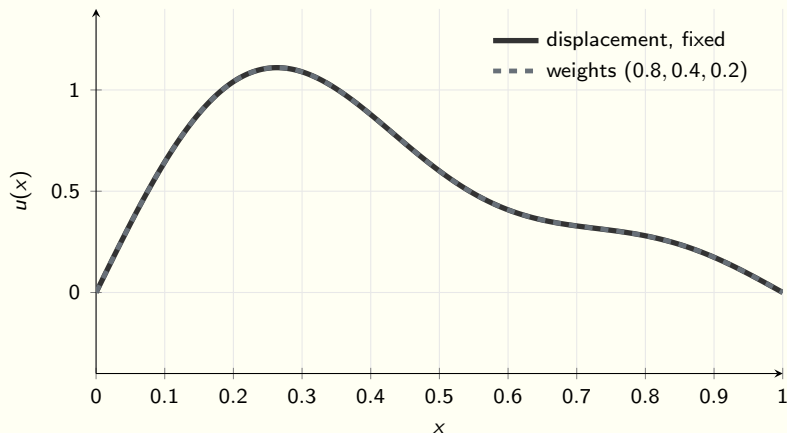
the family

$$u = a_1 \sin(\pi x) + a_2 \sin(2\pi x) + a_3 \sin(3\pi x)$$

one coefficient changes per panel

$(-0.4, 0.4, 0.2),$
 $(0.8, 0.9, 0.2),$
 $(0.8, 0.4, 0.7)$

Matching the displacement with three weights

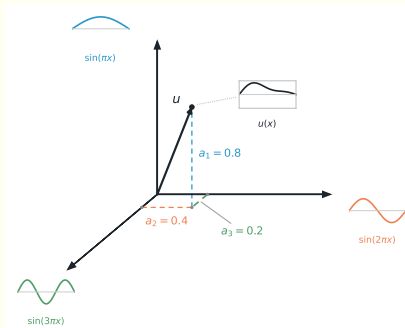
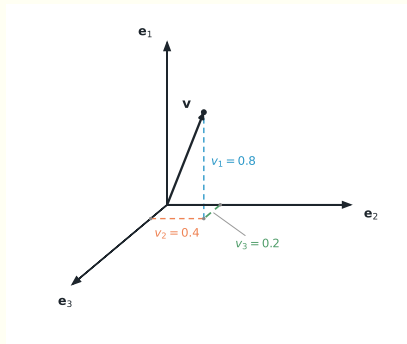


relative sup, L^2 , slope
errors
0, 0, 0

solve,
 $a_k = 2 \int_0^1 u \sin(k\pi x) dx$
(0.800, 0.400, 0.200)

[Try it live](#)

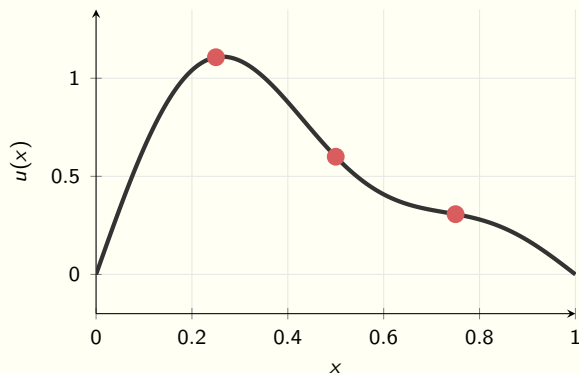
The coefficient list as coordinates in a vector space



$$u + v = \sum_{i=1}^3 (a_i + b_i) \phi_i,$$

$$\alpha u = \sum_{i=1}^3 (\alpha a_i) \phi_i$$

Recovering the coefficients from three readings



$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} & 1 & 1/\sqrt{2} \\ 1 & 0 & -1 \\ 1/\sqrt{2} & -1 & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$$

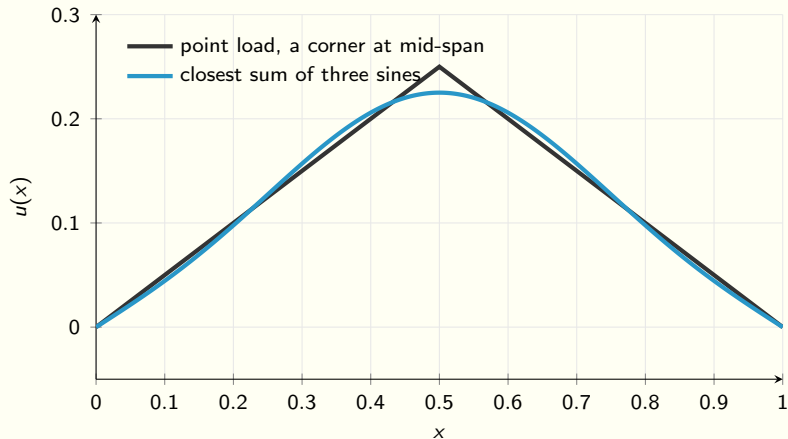
mid-span alone, $y_2 = u(\frac{1}{2})$
 $a_1 - a_3 = 0.6$, **nothing about a_2**

quarter spans, y_1, y_3
1.107 and 0.307

solved

$$\begin{aligned} a_2 &= \frac{y_1 - y_3}{2} = 0.4, \\ a_1 + a_3 &= \frac{y_1 + y_3}{\sqrt{2}} = 1, \\ \text{so } &(0.8, 0.4, 0.2) \end{aligned}$$

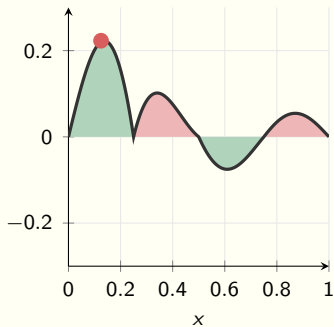
Approximating a function outside the span



closest three sines, relative
 L^2 error
4.8 percent

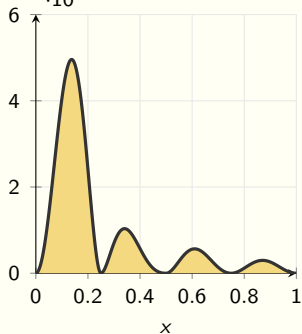
Measuring value and slope errors

error $e = u - v$



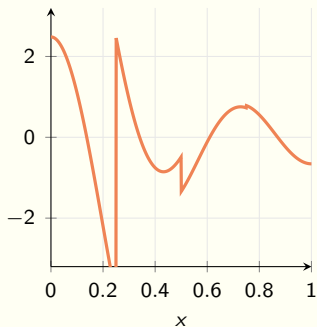
largest gap, sup norm
0.22

squared error e^2
 $\cdot 10^{-2}$



root of the area, L^2 norm
0.093

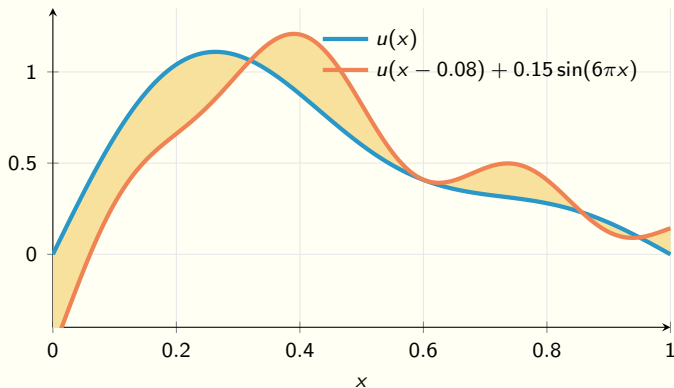
slope error e'



root-mean-square slope,
 H^1 seminorm
1.20

[Try it live](#)

The L^2 distance



$$\|u - g\|_{L^2} = \left(\int_0^1 (u - g)^2 dx \right)^{1/2}$$

L^2 distance

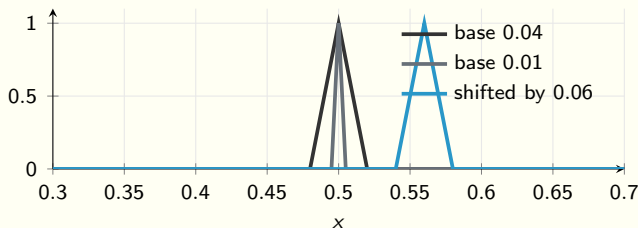
0.246

signed integral of the gap

0.0275

[norms notebook](#)

The sup norm, L^2 norm, and slope seminorm can disagree



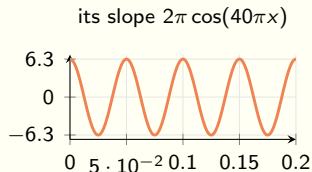
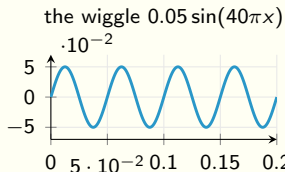
spike, base 0.04
sup 1, L^2 0.115,
energy 50

spike, base 0.01
sup 1, L^2 0.058,
energy 200

L^2 distance, spike to
shifted spike
 $\sqrt{2} \times 0.115 = 0.163$

wiggle $0.05 \sin(40\pi x)$
sup 0.05, L^2 0.035,
slope seminorm 4.44

[Try it live](#)



Definition of a norm and a seminorm

Norm

For functions u, v in a real vector space and a scalar α , a norm satisfies

$$\begin{aligned}\|u\| &\geq 0, & \|u\| = 0 &\iff u = 0, \\ \|\alpha u\| &= |\alpha| \|u\|, & \|u + v\| &\leq \|u\| + \|v\|.\end{aligned}$$

Seminorm

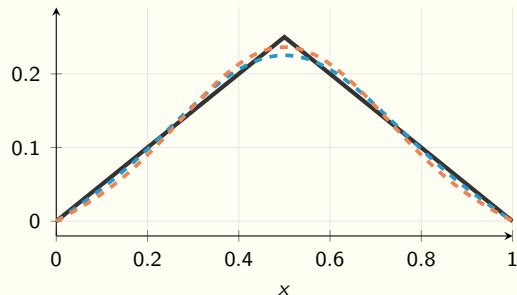
The slope seminorm vanishes on every constant function.

$$|c|_{H^1} = \|c'\|_{L^2} = 0.$$

Pinning excludes nonzero constants, so this seminorm is a norm on pinned finite-energy displacements.

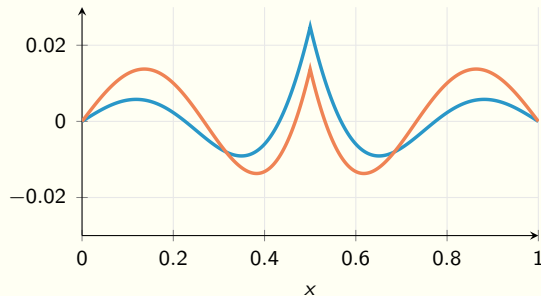
Closest three-sine sums to the corner in the L^2 and sup norms

the corner (black), closest in L^2 (blue) and in sup (orange)



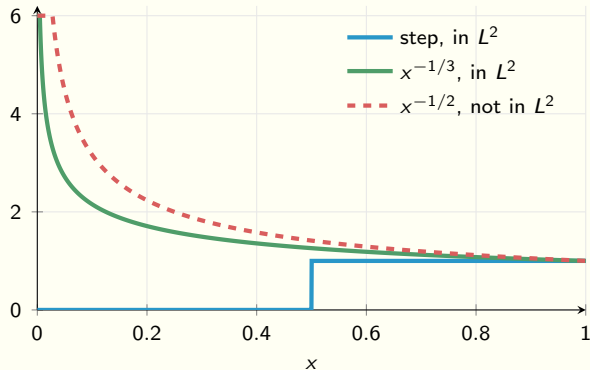
L^2 -closest,
(0.2026, 0, -0.0225)
**sup 9.9, L^2 4.8, slope
31.5 percent**

the gap $u - v$



sup-closest,
(0.2045, 0, -0.0318)
**sup 5.5, L^2 6.7, slope
33.9 percent**

The space L^2 of square-integrable functions



$$\|f\|_{L^2}^2 = \int_0^1 f(x)^2 dx < \infty$$

$$\langle f, g \rangle = \int_0^1 fg dx$$

$\int_0^1 (x^{-1/3})^2 dx$
3, finite although
 $x^{-1/3}$ is unbounded

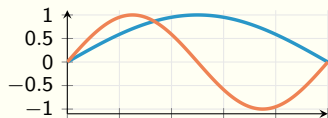
$\int_0^1 (x^{-1/2})^2 dx =$
 $\int_0^1 x^{-1} dx$
diverges

two functions equal except
at one point
 L^2 distance 0

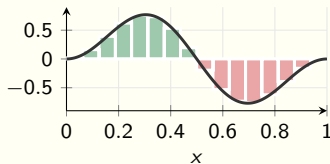
The inner product as signed overlap

$f = \sin(\pi x)$ blue, g orange, and below them fg , green where positive and red where negative

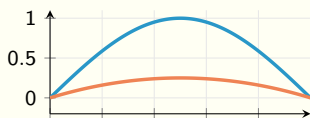
$g = \sin(2\pi x)$



fg , net area 0



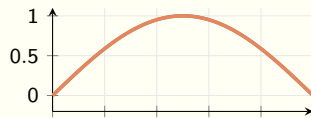
$g = x(1 - x)$



fg , net area 0.129



$g = f$

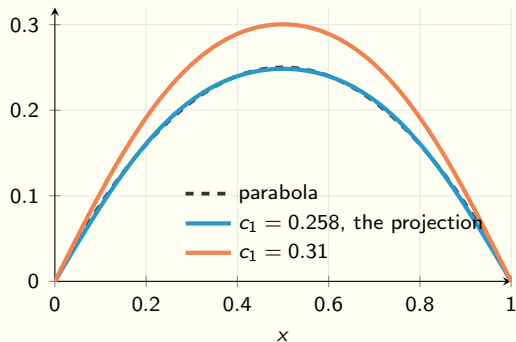


fg , net area 0.500



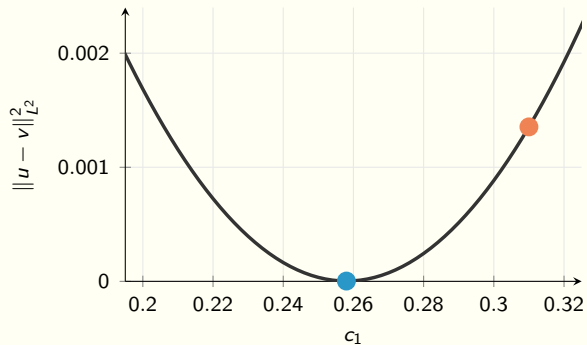
$$\langle f, g \rangle = \int_0^1 f(x) g(x) dx \approx \frac{1}{16} \sum_{i=1}^{16} f(x_i) g(x_i)$$

Minimizing squared error via orthogonal projection



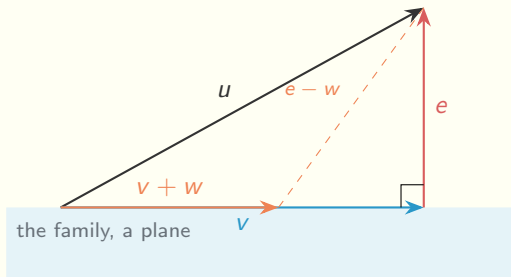
$c_1 = 8/\pi^3 = 0.258$
error 0.87 percent

$c_1 = 0.31$, twenty percent high
error norm 23 times larger



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Uniqueness of the projection and the Pythagorean identity



$$\|e - w\|^2 = \|e\|^2 + \|w\|^2$$
$$\|u\|^2 = \|v\|^2 + \|e\|^2, \quad 0.033333 = 0.033331 + 2$$

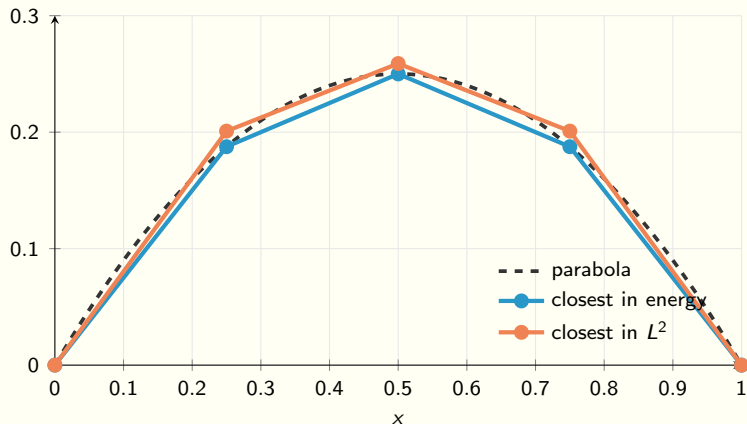
The projection theorem

Projection theorem

For each target in an inner-product space, a finite-dimensional linear subspace has a unique closest member. Its error is orthogonal to that subspace.

$$\langle u - v, \phi_j \rangle = 0 \quad \text{for every shape } \phi_j, \quad c_k = \frac{\langle u, \phi_k \rangle}{\langle \phi_k, \phi_k \rangle} \text{ when the shapes are orthogonal.}$$

Projection in the L^2 and energy inner products



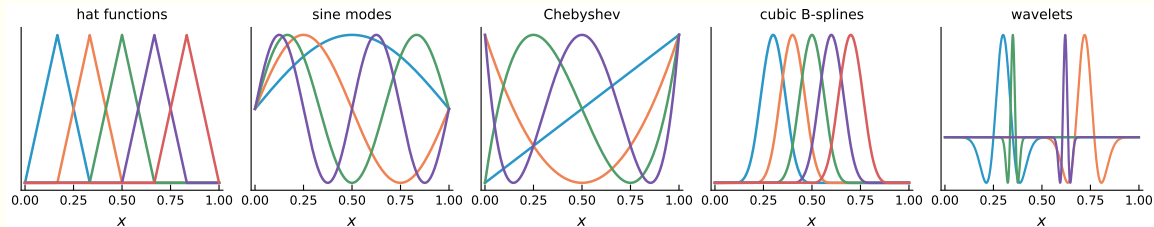
closest in energy, heights
0.1875, 0.25, 0.1875

closest in L^2 , heights
0.2009, 0.2589, 0.2009

L^2 error, L^2 against energy
3.3 against 6.3 percent

relative slope error, L^2
against energy
26.0 against 25.0 percent

Five fixed families of shapes



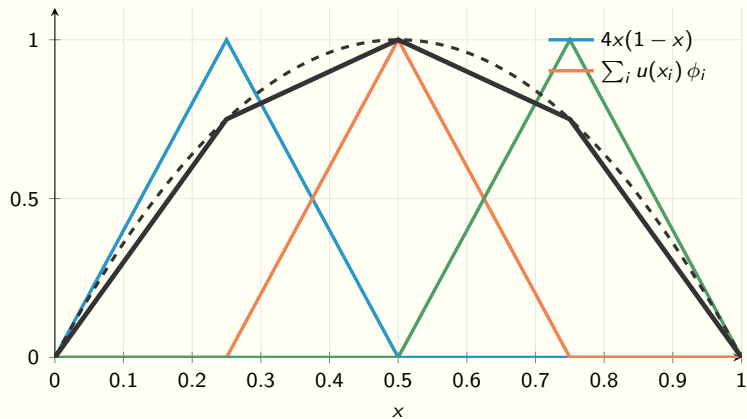
local
hats, one per node

global
**sine modes,
Chebyshev
polynomials**

smooth and local
cubic B-splines

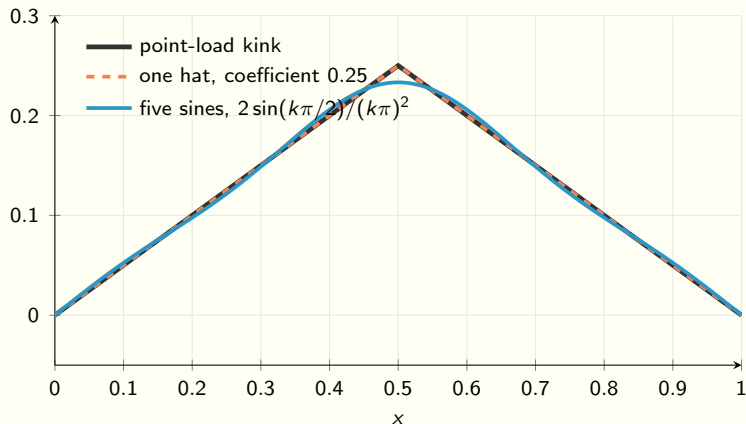
local in position and scale
wavelets

Piecewise-linear interpolation as a hat-function expansion



$$u_h(x) = \sum_i u(x_i) \phi_i(x)$$

Representing a kink in the hat basis and the sine basis



one hat on the corner

one number, 0.25, no error

three hats, five stations

0.125, 0.25, 0.125, no error

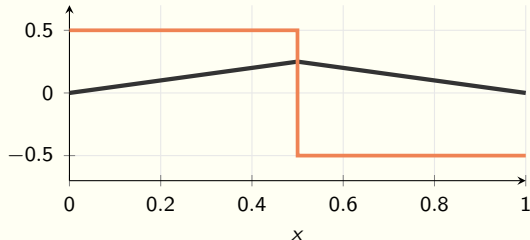
relative L^2 error, three and thirty-three sines

4.8 and 0.20 percent

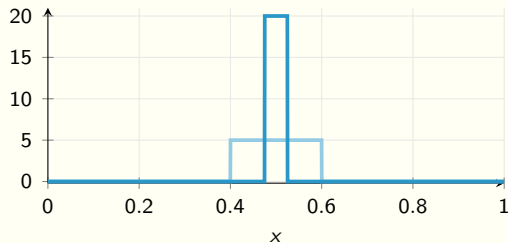
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The space H^1 of finite-energy functions

the corner $\min(x, 1-x)/2$ and its slope, in H_0^1



ramps toward a step, and their slopes $1/w$



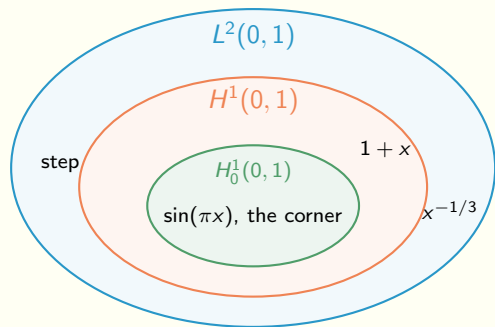
$$\|u\|_{H^1}^2 = \int_0^1 u^2 dx + \int_0^1 (u')^2 dx,$$

$$\langle u, v \rangle_{H^1} = \int_0^1 (uv + u'v') dx$$

ramp of width w ,
 $\int (u')^2 dx$
 **$1/w$, so 5 at $w = 0.2$
and 20 at $w = 0.05$**

in one dimension
 $|u(y) - u(x)| \leq$
 $\sqrt{|y - x|} \|u'\|_{L^2}$

The nesting $H_0^1 \subset H^1 \subset L^2$



$$1+x$$

slope 1, but $u(0) = 1$

the step

ramps toward it have

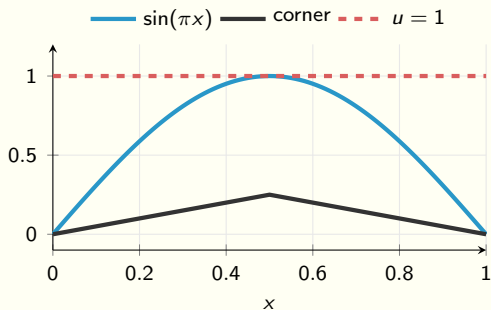
$$\int (u')^2 dx = 1/w$$

$$x^{-1/3}$$

$$\int_0^1 (u')^2 dx =$$

$$\int_0^1 \frac{1}{9} x^{-8/3} dx \text{ **diverges**}$$

The space H_0^1 and the Poincaré inequality



$\|u\|_{L^2}/|u|_{H^1}$ for $\sin(\pi x)$
 $0.3183 = 1/\pi$,
equality

for the corner
 $0.1443/0.5 = 0.2887$

$u = 1$, in H^1 only
 $|u|_{H^1} = 0$, **the bound fails**

$$H_0^1(0, 1) = \{ u \in H^1(0, 1) : u(0) = u(1) = 0 \}$$

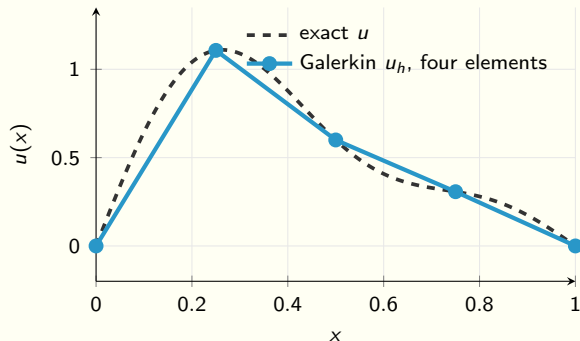
$$\|u\|_{L^2} \leq \frac{1}{\pi} |u|_{H^1} \quad \text{for } u \in H_0^1(0, 1)$$

$$|u|_{H^1}^2 \leq \|u\|_{H^1}^2 \leq \left(1 + \frac{1}{\pi^2}\right) |u|_{H^1}^2 \quad \text{on } H_0^1$$

The spaces L^2 , H^1 , and H_0^1 compared

space	$u \in L^2$	$\nabla u \in L^2$	inner product	boundary values	role in $-\Delta u = f$
$L^2(\Omega)$	yes	not required	$\int_{\Omega} uv \, dx$	no general trace or point evaluation	square-integrable loads and value misfits
$H^1(\Omega)$	yes	yes	$\int_{\Omega} (uv + \nabla u \cdot \nabla v) \, dx$	trace exists, none imposed	free (Neumann) boundaries, imposed through the weak form
$H_0^1(\Omega)$	yes	yes	$\int_{\Omega} \nabla u \cdot \nabla v \, dx$	$u = 0$ on $\partial\Omega$	trial and test space for fixed (Dirichlet) boundaries

Galerkin's method with hat functions



$$\int_0^1 u_h' \phi_i' dx = \int_0^1 f \phi_i dx \quad \text{for every hat } \phi_i$$

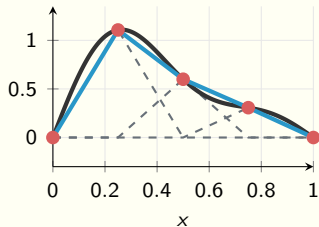
$$\mathbf{K}\mathbf{u} = \mathbf{F}, \quad K_{ii} = \frac{2}{h}, \quad K_{i,i\pm 1} = -\frac{1}{h}$$

largest nodal deviation, 4
to 16 elements
 4.4×10^{-16}

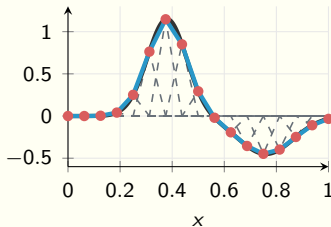
error slopes on log axes, L^2
and energy
2.00 and 1.00

Convergence of hat interpolation for continuous and discontinuous functions

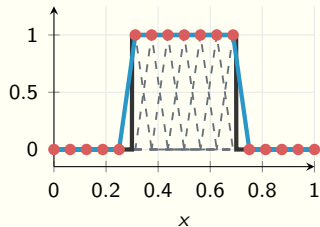
the string, five stations



a bump, seventeen stations



a square pulse, seventeen stations



$$u_h(x) = \sum_i u(x_i) \phi_i(x), \quad \phi_i(x_j) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

string, largest gap at 5, 17, 33 stations

20.1, 1.6, 0.4 percent

bump, largest gap at 9, 33, 65 stations

13.4, 2.1, 0.5 percent

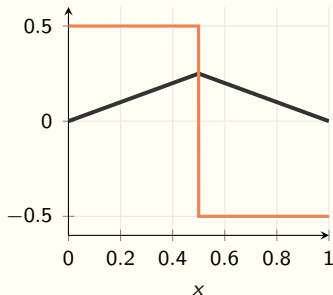
pulse, largest gap and RMS gap at 17

80 and 23.3 percent

[Try it live](#)

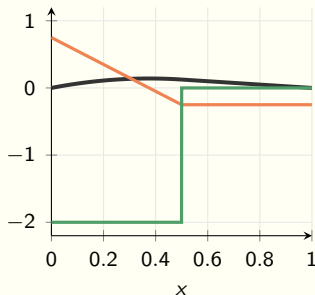
Regularity from the load to the displacement

point load, $u \in H_0^1$



point load
slope jumps,
 $c_k \sim 1/k^2$

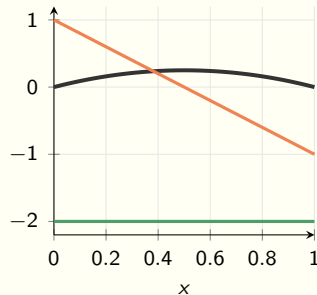
step load, $u \in H^2$



u black, u' orange, u'' green

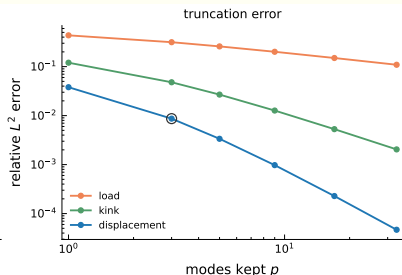
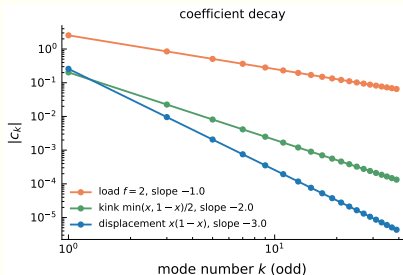
step load in L^2
curvature jumps,
 $u'' \in L^2$

uniform load, $u = x(1-x)$



uniform load
smooth inside,
 $c_k \sim 1/k^3$

Decay rates of sine coefficients

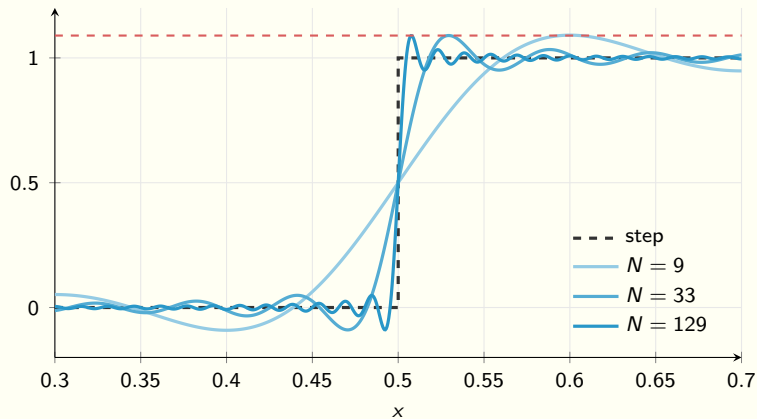


load $f=2$, odd k
 $|c_k| = 8/(k\pi)$, slope -1.0

kink, odd k
 $|c_k| = 2/(k\pi)^2$, slope -2.0

parabola, odd k
 $|c_k| = 8/(k\pi)^3$, slope -3.0

The Gibbs phenomenon at a jump



overshoot, $N = 9$

0.091

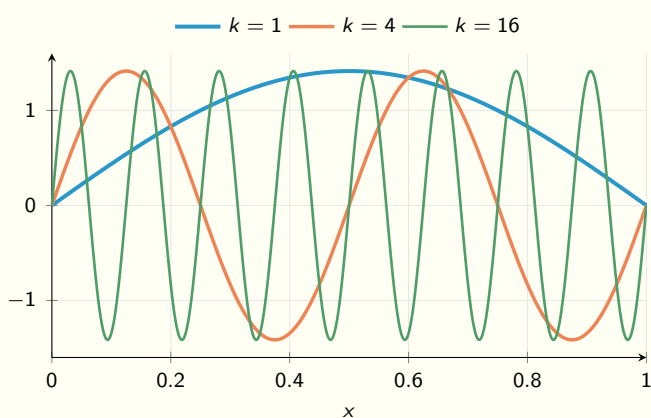
overshoot, $N = 33$

0.090

overshoot, $N = 129$

0.089, **toward** 0.0895

Equal L^2 norm, increasing slope seminorm



$$\phi_k(x) = \sqrt{2} \sin(k\pi x)$$

$$\|\phi_k\|_{L^2} = 1, \quad |\phi_k|_{H^1} = k\pi$$

$|\phi_k|_{H^1}$ at $k = 1$
3.14

$k = 4$
12.57

$k = 16$
50.27

distance between two
different modes
 $\sqrt{2} = 1.414$

Compactness from bounded energy

$$u = \sum_{k \geq 1} c_k \sqrt{2} \sin(k\pi x), \quad |u|_{H^1}^2 = \sum_{k \geq 1} (k\pi)^2 c_k^2$$

$$|u|_{H^1} \leq M \implies \left(\sum_{k > K} c_k^2 \right)^{1/2} \leq \frac{M}{(K+1)\pi}$$

unit ball of L^2

**the modes stay $\sqrt{2}$
apart, no
subsequence
converges**

$M = 10$, tail beyond $K = 4$

at most 0.637

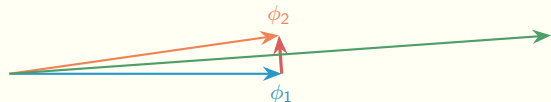
$M = 10$, tail beyond
 $K = 16$

at most 0.187

Rellich, on a bounded
interval

**a bounded sequence
in H_0^1 has an
 L^2 -convergent
subsequence**

Conditioning of the Gram matrix for nearly parallel shapes



$\phi_2 - \phi_1$, length 0.14 $\phi_1 + \phi_2$, length 1.995

$$\mathbf{G} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \quad \rho = 0.99, \quad \text{cond } \mathbf{G} = \frac{1 + \rho}{1 - \rho}$$

add $0.1(\phi_1 - \phi_2)$

target moves 0.014,
coefficients 0.1 and
-0.1

add $0.1(\phi_1 + \phi_2)$

target moves 0.2,
fourteen times more

$\sqrt{199}$

14.1

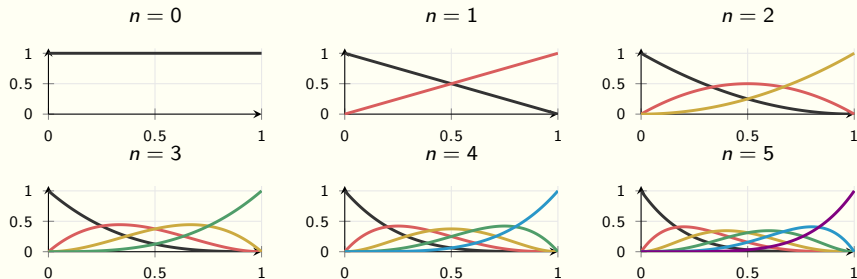
neighboring hats

$\rho = 1/4$

five Gaussian bumps a
third of a width apart

cond $\mathbf{G} = 2.4$ million

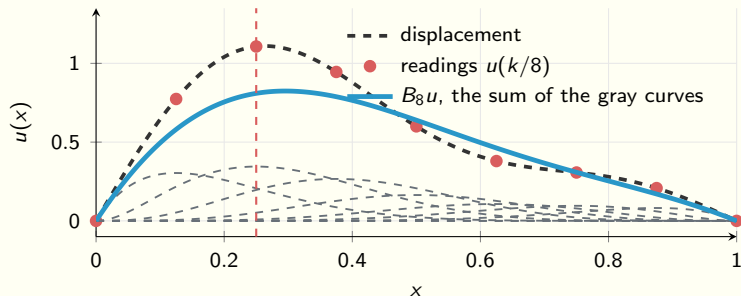
Bernstein basis polynomials of degree 0 to 5



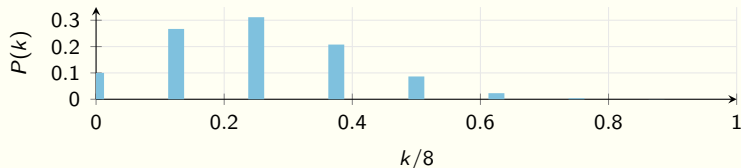
$$B_{i,n}(t) = \binom{n}{i} t^i (1-t)^{n-i}, \quad \sum_{i=0}^n B_{i,n}(t) = 1, \quad B_{i,n}(t) \geq 0$$

$B_{i,n}(t)$ is the probability of exactly i heads in n flips of a coin that lands heads with probability t

Bernstein polynomials for the string displacement



probability of k heads in 8 coins, $x = 0.25$



$$B_n u(x) = \sum_{k=0}^n u\left(\frac{k}{n}\right) b_{k,n}(x)$$

$$b_{k,n}(x) = \binom{n}{k} x^k (1-x)^{n-k}$$

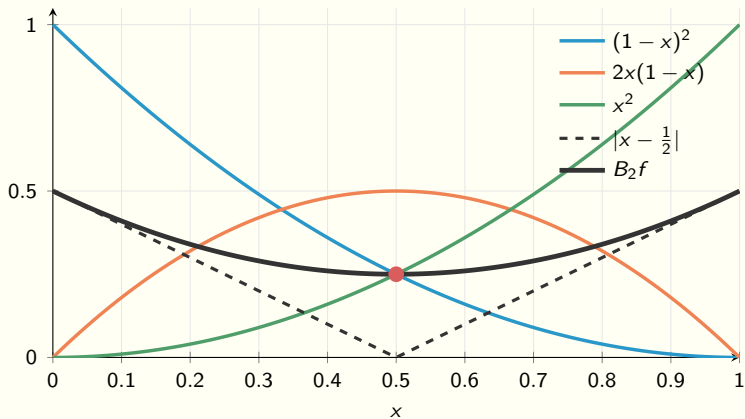
$B_8 u(0.25)$, the weighted average

0.810 against
 $u(0.25) = 1.107$

largest gap, $n = 8, 32, 128$
0.30, 0.091, 0.024

Voronovskaya, large n
 $3.18/n$

Approximating a corner with Bernstein polynomials



Bernstein basis weights at mid-span

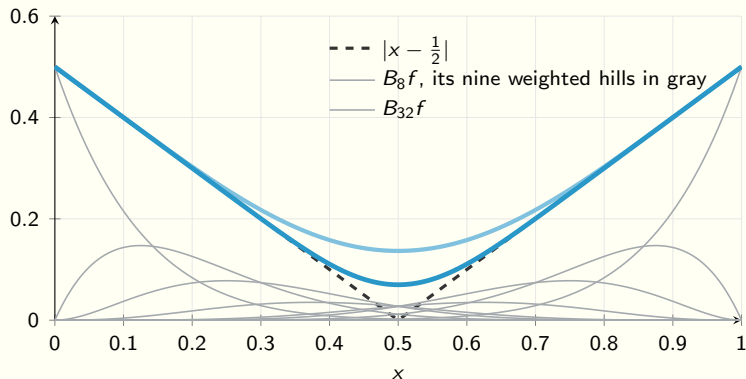
$\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$

readings of $|x - \frac{1}{2}|$ at $0, \frac{1}{2}, 1$

$\frac{1}{2}, 0, \frac{1}{2}$

$B_2 f$ at the corner
 $\frac{1}{4}$ where $f = 0$

Convergence rate of Bernstein polynomials



$$B_n f(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}$$

gap at the corner, $n = 2$
0.250

gap at the corner, $n = 8$
0.137, **predicted**
0.141

gap at the corner, $n = 32$
0.070, **predicted**
0.071

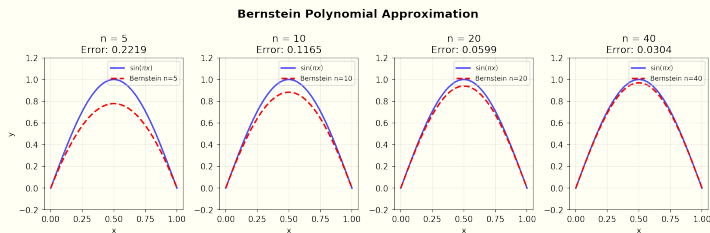
prediction
 $0.4/\sqrt{n}$

[Try it live](#)

The Weierstrass approximation theorem

Theorem (Weierstrass, 1885)

For every continuous f on $[a, b]$ and every $\epsilon > 0$ there is a polynomial p with $|f(x) - p(x)| < \epsilon$ for every x in $[a, b]$.



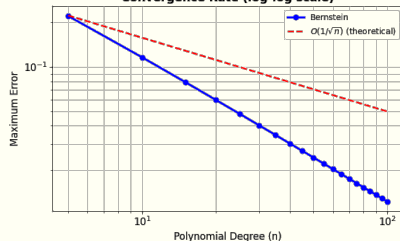
Bernstein $B_n f$ for $\sin(\pi x)$,
largest gap
0.222, 0.117, 0.060,
0.030 at
 $n = 5, 10, 20, 40$

what the theorem does not
say
**the degree, the
weights, or the
slopes**

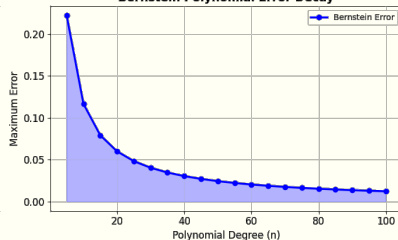
Convergence of Bernstein polynomials for a smooth function

Bernstein Polynomial Convergence: Proof of Weierstrass Theorem

Convergence Rate (log-log scale)



Bernstein Polynomial Error Decay



target
 $\sin(\pi x)$

largest gap, $n = 5$ and
 $n = 100$

0.22 and 0.013

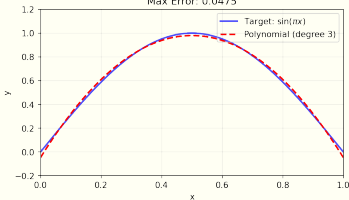
Voronovskaya at mid-span,
 $\pi^2/(8n)$

0.25 and 0.012

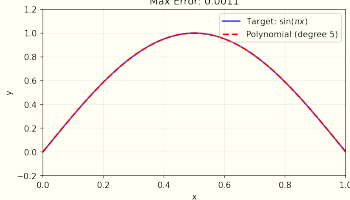
Least-squares polynomial approximation of $\sin(\pi x)$

Polynomial Approximation of $\sin(\pi x)$ (Weierstrass)

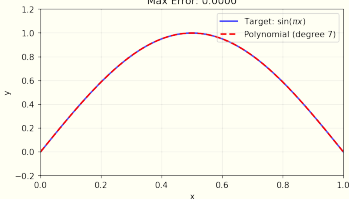
Degree 3 Approximation
Max Error: 0.0475



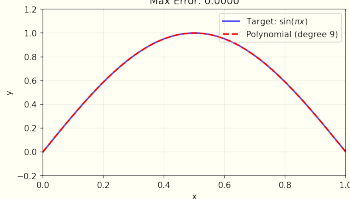
Degree 5 Approximation
Max Error: 0.0011



Degree 7 Approximation
Max Error: 0.0000



Degree 9 Approximation
Max Error: 0.0000



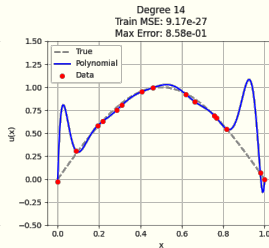
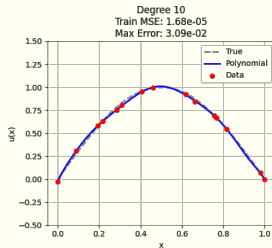
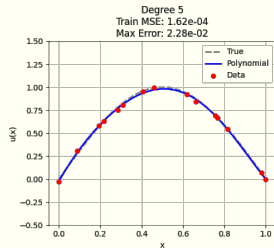
degree 3, largest gap
0.0475

degree 5, largest gap
0.0011

degrees 7 and 9, largest
gap
below 0.00005

High-degree interpolation of noisy data

Runge's Phenomenon: Higher Degree $\rightarrow$ Worse Oscillations



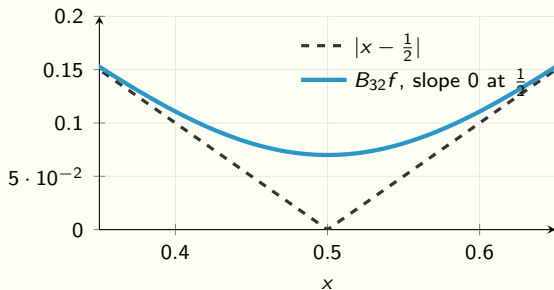
degree 5, least squares
largest gap 0.023

degree 10, least squares
largest gap 0.031

degree 14, through all
fifteen noisy readings
**training mean
squared error**
 9×10^{-27} , **largest
gap 0.86**

What the Weierstrass theorem does not say

$B_{32}f$ beside the corner



Bernstein sup error for this corner

falls like $1/\sqrt{n}$

best degree- n sup error for this corner

falls like $1/n$

the theorem names

no polynomial, no degree, no slope

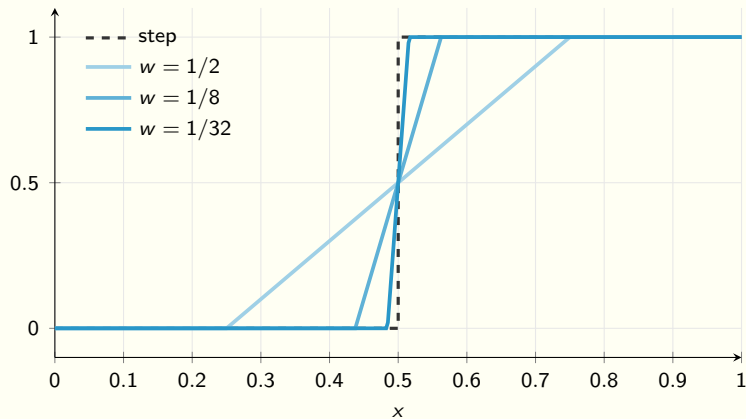
Polynomial density through vanishing moments

moments forced to zero	response to $\sin(3\pi x)$
0	0.635
2	0.565
4	0.298
8	1.2×10^{-3}
16	3.0×10^{-9}
32	6.9×10^{-16}

$$\int_0^1 x^k d\mu(x) = 0 \quad \text{for every } k$$
$$\implies \int_0^1 f d\mu = 0 \quad \text{for every continuous } f$$

density criterion
**all polynomial
moments zero
implies the finite
signed measure is
zero**

Cauchy sequences, completeness, and Hilbert spaces



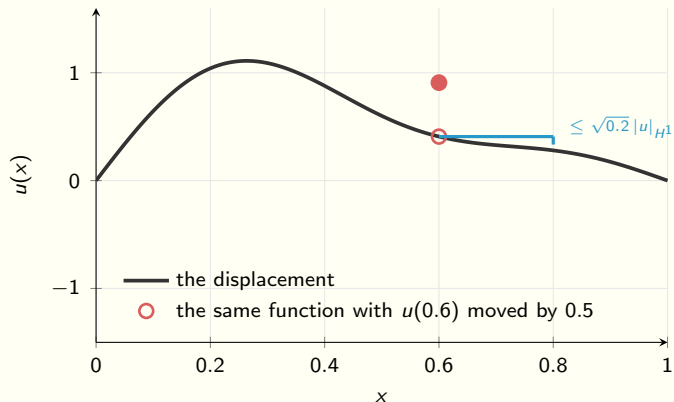
L^2 distance to the step,
 $\sqrt{w/12}$
0.204 at $w = 1/2$

$w = 1/8$
0.102

$w = 1/32$
0.051

$\|f_8 - f_{32}\|_{L^2}$
0.077

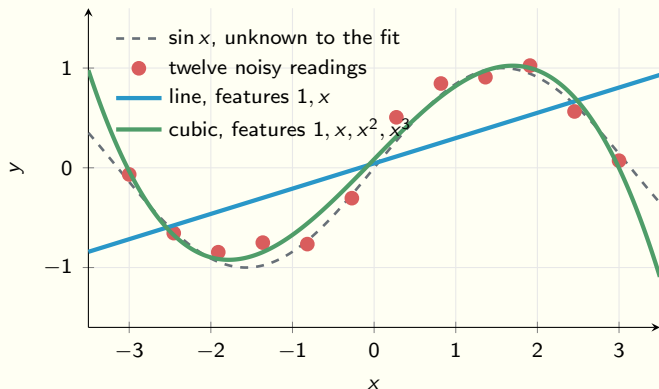
Point evaluation in L^2 and H^1



L^2 distance between the two
0

$$|u(y) - u(x)| = \left| \int_x^y u'(s) ds \right| \leq \sqrt{y - x}$$

Linear regression by least squares



$$\hat{y}(x) = \sum_j w_j \phi_j(x), \quad \min_{\mathbf{w}} \|\Phi \mathbf{w} - \mathbf{y}\|^2$$

$$\Phi^T \Phi \mathbf{w} = \Phi^T \mathbf{y}$$

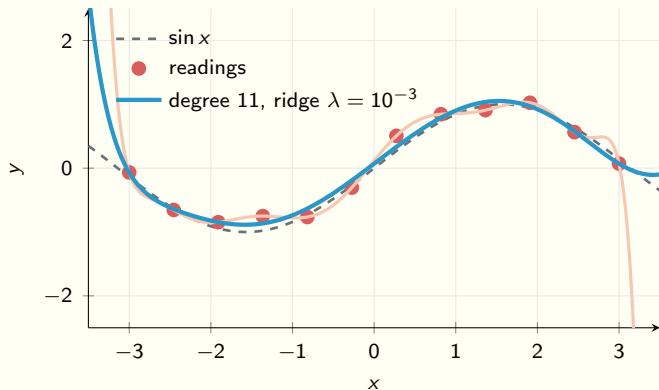
line, RMSE at the data
and against $\sin x$

0.484 and 0.439

cubic, the same two errors

0.119 and 0.085

Ridge regression on many features



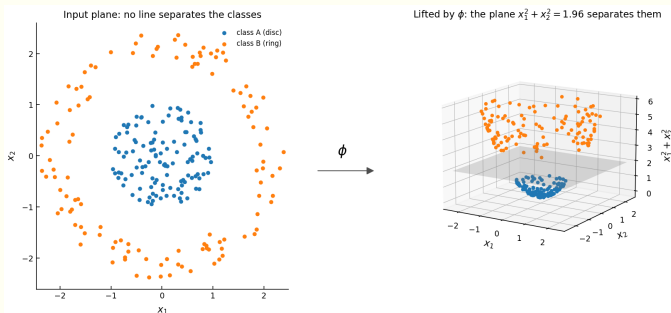
$$\min_{\mathbf{w}} \|\Phi \mathbf{w} - \mathbf{y}\|^2 + \lambda \|\mathbf{w}\|^2$$

$$(\Phi^T \Phi + \lambda \mathbf{I}) \mathbf{w} = \Phi^T \mathbf{y}$$

degree 11, least squares
error at the data 0,
largest weight 284,
swings to 29.5

degree 11, ridge
error against $\sin x$
0.073, largest weight
4.3

A kernel is an inner product of features



$$\phi(x_1, x_2) = (x_1, x_2, x_1^2 + x_2^2)$$

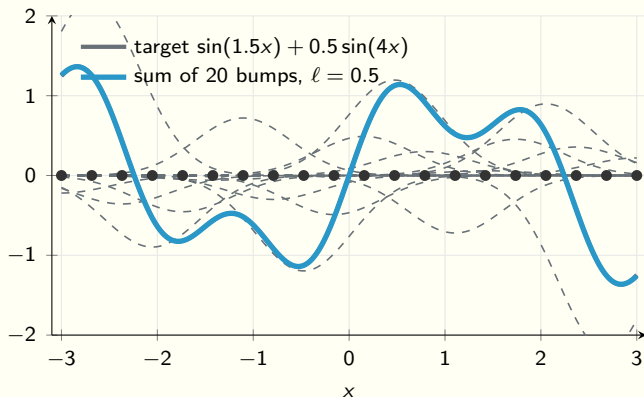
$$k(\mathbf{x}, \mathbf{x}') = \phi(\mathbf{x}) \cdot \phi(\mathbf{x}') = \mathbf{x} \cdot \mathbf{x}' + |\mathbf{x}|^2 |\mathbf{x}'|^2$$

a plane in feature space

$x_3 = 1.96$ separates
disc from ring, no
errors

$\mathbf{x} = (1, 0)$, $\mathbf{x}' = (0.5, 0.5)$
both routes give
1.000

Regression on Gaussian bump features



$$\hat{f}(x) = \sum_{j=1}^m w_j e^{-(x-c_j)^2/2\ell^2}$$

5 bumps, $\ell = 0.5$

RMSE 0.442

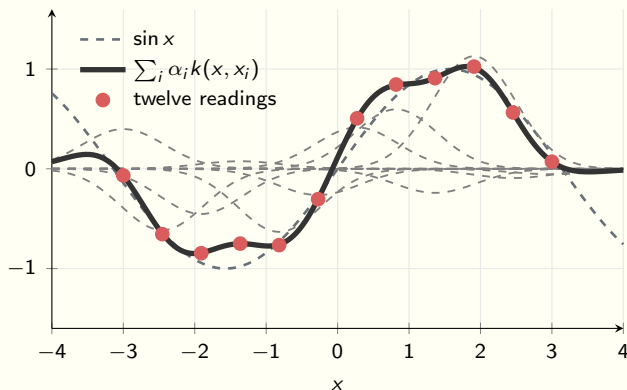
20 bumps, $\ell = 0.5$

RMSE 0.002

20 bumps, narrow $\ell = 0.1$
and wide $\ell = 2$

0.191 with gaps,
0.316 blurred

Kernel ridge regression with copies at the data



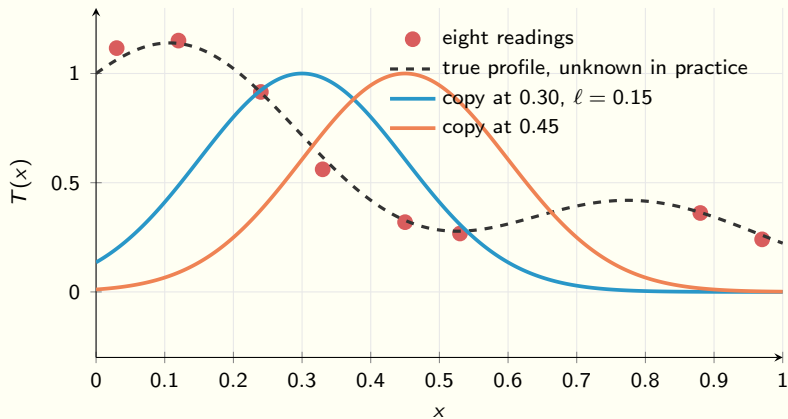
$$\hat{f}(x) = \sum_{i=1}^{12} \alpha_i k(x, x_i), \quad (\mathbf{K} + \lambda \mathbf{I}) \boldsymbol{\alpha} = \mathbf{y}$$

weights, $\ell = 1/\sqrt{3}$,
 $\lambda = 0.01$

from -0.632 to 1.129

RMSE at the readings and
against $\sin x$
0.0051 and 0.113

Kernels as similarity rules for scattered readings

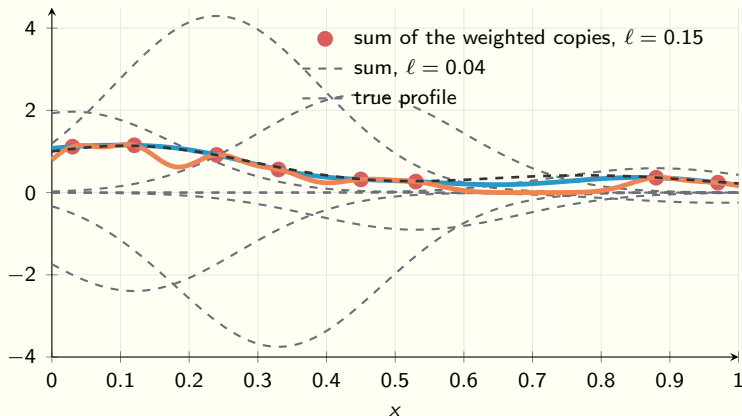


$$k(x, x') = e^{-(x-x')^2/2\ell^2}$$

$k(0.30, 0.45)$, one ℓ apart
0.61

k at 3ℓ apart
0.011

Reconstructing a profile with weighted kernel copies



$$\mathbf{K}\alpha = \mathbf{y}, \quad K_{ij} = k(x_i, x_j)$$

weights at $\ell = 0.15$

alternate in sign, up to 4.30 and -3.75

two equal readings one ℓ apart

$1/(1 + \rho) = 0.62$ **each**

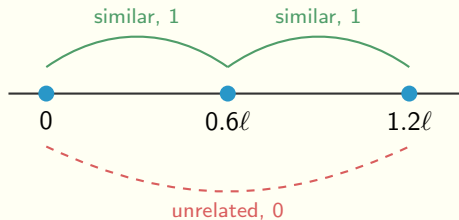
at $x = 0.7$, $\ell = 0.15$

0.216 against the reference 0.393

sum at $x = 0.7$, $\ell = 0.04$

0.000 against the true 0.393

Positive definiteness of a kernel

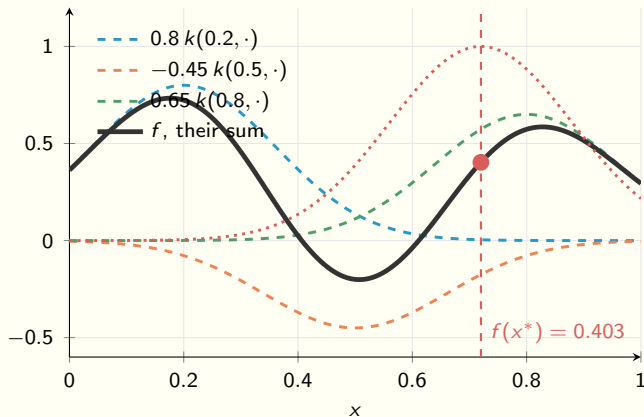


$$\mathbf{K} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad \lambda = 1 - \sqrt{2}, 1, 1 + \sqrt{2}$$

$$\mathbf{w} = (1, -\sqrt{2}, 1), \quad \mathbf{w}^\top \mathbf{K} \mathbf{w} = -1.66$$

$$k(x, x') = \phi(x)^\top \phi(x') \Rightarrow \mathbf{w}^\top \mathbf{K} \mathbf{w} = \left\| \sum_i w_i \phi(x_i) \right\|^2$$

The reproducing kernel Hilbert space



$$\langle k(x_1, \cdot), k(x_2, \cdot) \rangle_k = k(x_1, x_2)$$

$$\|f\|_k^2 = \alpha^\top K \alpha, \ell = 0.16$$

1.041

$$f(x) = \langle f, k(x, \cdot) \rangle_k$$

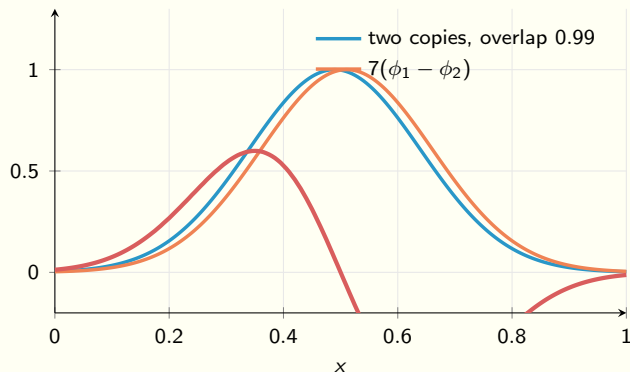
$f(0.72)$, read as an inner product

0.403, **bound** 1.02

slope bound

$$|f'(x)| \leq \|f\|_k / \ell$$

Noise amplification in kernel interpolation



eigenvalues of $\mathbf{K}$, overlap 0.99

1.99 along $(1, 1)$, 0.01 along $(1, -1)$

readings $(1, 1)$

weights near $\frac{1}{2}$ each

noise $(\varepsilon, -\varepsilon)$

weights $\pm\varepsilon/0.01$, a hundred times the noise

RKHS norm of the perturbation

$14.1|\varepsilon|$, distinct from pointwise error

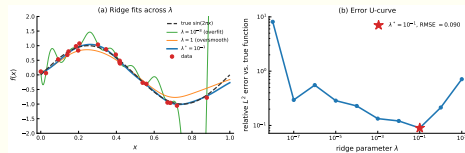
Kernel ridge regression and the choice of λ

$$\min_{f \in \mathcal{H}_k} \sum_{i=1}^n (f(x_i) - y_i)^2 + \lambda \|f\|_k^2$$

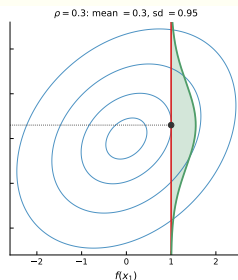
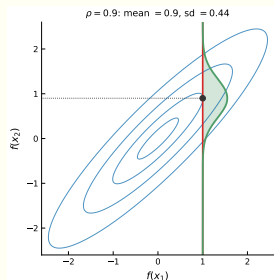
$$(\mathbf{K} + \lambda \mathbf{I})\alpha = \mathbf{y}$$

$\lambda = 0.1$, small divisor
 $0.01 \rightarrow 0.11$

amplification
100 **to** 9.1



Conditioning a Gaussian on an observation



$$f(x_1) = Z_1, \quad f(x_2) = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$$

$$f(x_2) \mid f(x_1) = y \sim \mathcal{N}(\rho y, 1 - \rho^2)$$

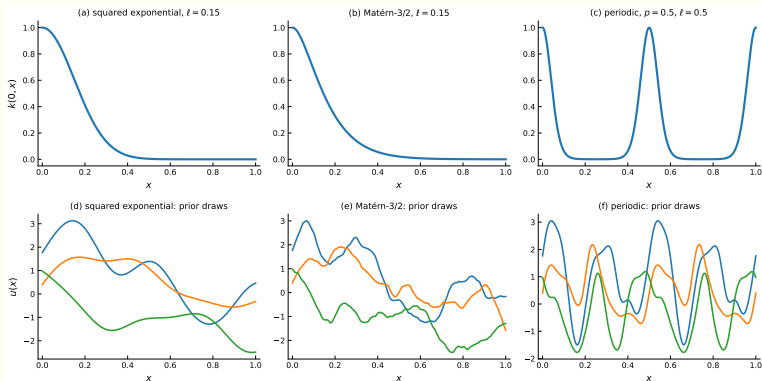
$$\rho = 0.9$$

**mean $0.9y$, standard
deviation 0.44**

$$\rho = 0.3$$

**mean $0.3y$, standard
deviation 0.95**

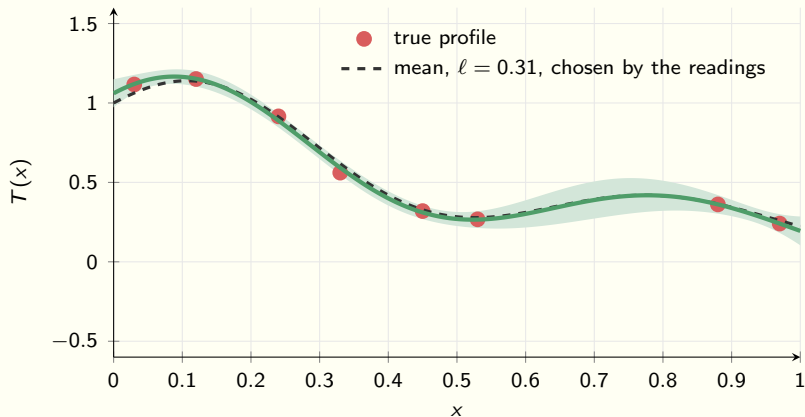
Prior draws from three kernels



mean absolute slope of the
draws
5.2, 8.9, and 20

Gaussian-process model
**the prior specifies a
mean and a
covariance kernel**

Gaussian-process reconstruction of the fin profile



mean

$$\mathbf{k}(x)^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{y}$$

variance

$$k(x, x) - \mathbf{k}(x)^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}(x)$$

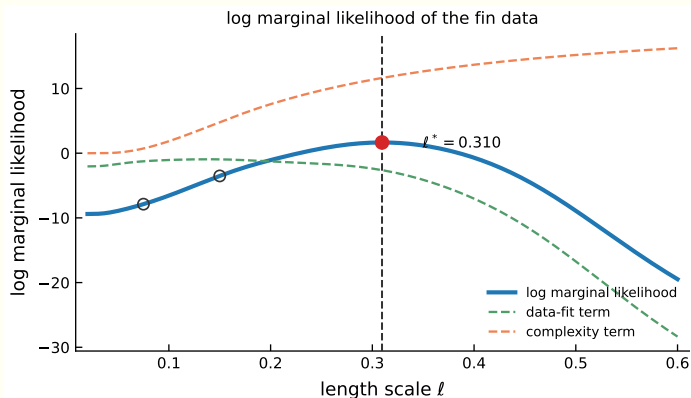
$\ell = 0.31$

**error 3.2 percent,
half-width in the gap
0.12**

value at $x = 0.7$

**0.389 against the
true 0.393**

Choosing the length scale by marginal likelihood



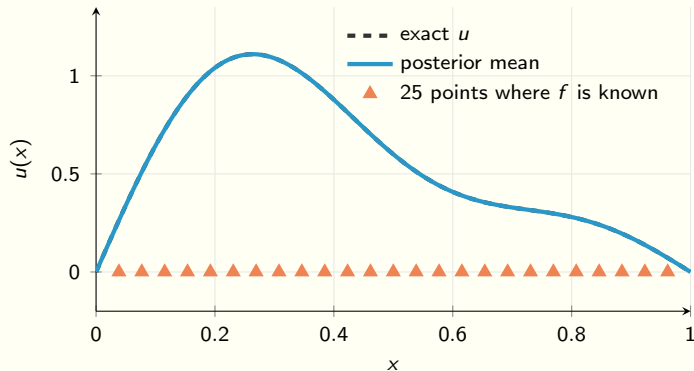
log marginal likelihood at
 $\ell = 0.31$

1.7, the maximum

at $\ell = 0.15$ and 0.075
-3.5 and -7.9

length-scale selection
**the posterior band
conditions on the
selected value**

Conditioning a kernel prior on $\mathcal{L}u = f$



$$\text{cov}(u, f) = \mathcal{L}_{x'} k$$
$$\text{cov}(f, f) = \mathcal{L}_x \mathcal{L}_{x'} k$$

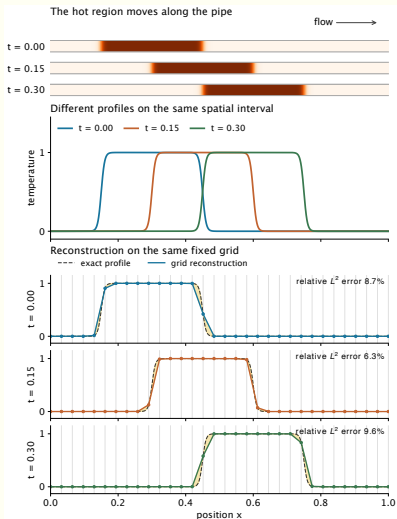
relative L^2 error of the
mean

$$3.2 \times 10^{-7}$$

condition number of the
observation covariance

$$1.4 \times 10^{12}$$

A moving front gives a family of functions



$$T(x, t) = T_0(x - ct), \quad \mathcal{S} = \{T_0(\cdot - ct) \mid 0 \leq t \leq 0.30\}$$

$$\hat{T}(x, t) = \sum_i T(x_i, t) \phi_i(x)$$

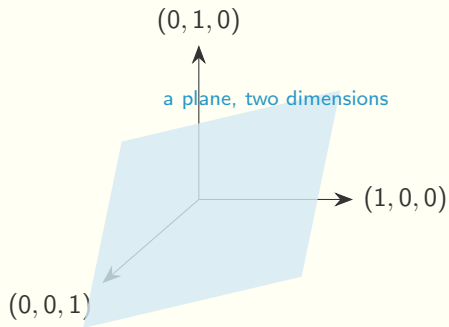
relative L^2 error, 32 nodes,
times 0, 0.15, 0.30

8.7, 6.3, 9.6 percent

edge width against node
spacing

0.01 against 1/31

The Kolmogorov n -width



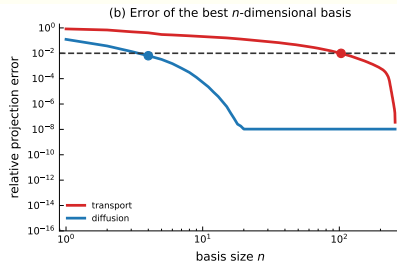
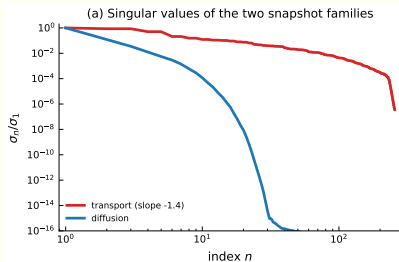
Kolmogorov n -width

$$d_n(\mathcal{M}) = \inf_{\dim V=n} \sup_{f \in \mathcal{M}} \inf_{v \in V} \|f - v\|$$

three unit spikes, squared
distances to any plane
sum to $3 - 2 = 1$

worst spike
at least
 $\sqrt{1/3} = 0.577$

Widths of diffusion and transport families



diffusing pulse, SVD, one percent relative Frobenius error

four modes

translating pulse, same measure

103 modes

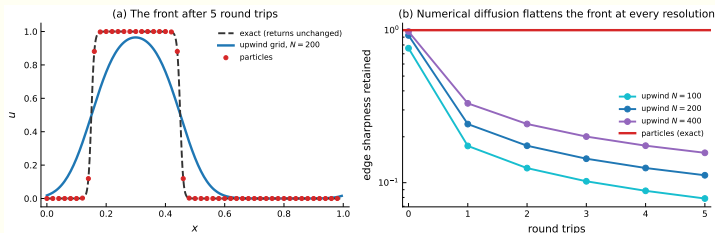
coefficient k under diffusion

shrinks by $e^{-(2\pi k)^2 t}$

coefficient k under translation

changes phase only

Particle positions and values under advection



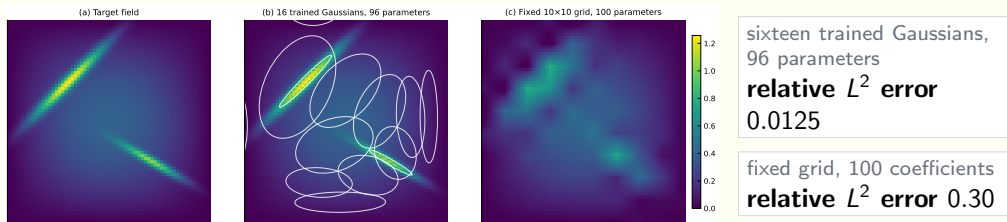
markers after five round trips

the front returns unchanged

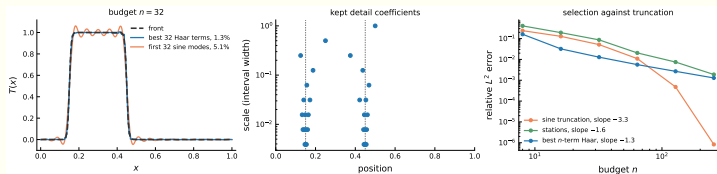
upwind grid at 100, 200, 400 points

keeps 8, 11, and 16 percent of the edge steepness

Gaussian splats with trainable centers and shapes



Selecting coefficients from a fixed dictionary

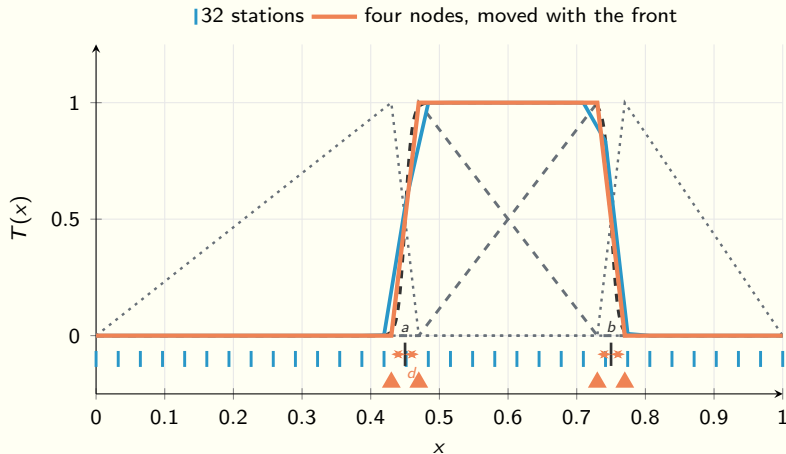


relative L^2 error, $n = 16$,
sines, stations, Haar
12.8, 19.3, 3.2
percent

relative L^2 error, $n = 32$,
Haar against sines
1.3 against 5.1
percent

smaller relative L^2 error for
sines
near $n = 128$

Movable hat nodes on a traveling front



nodes

$a - d$, $a + d$, $b - d$,
 $b + d$, with $d = 0.02$

32 stations, front shifted
by 0.3

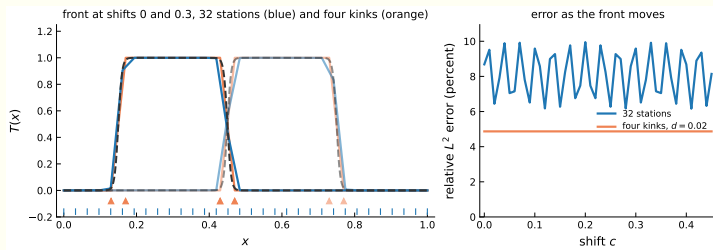
9.6 percent

four nodes, moved with
the front

4.9 percent

[Try it live](#)

Fixed nodes and moving nodes



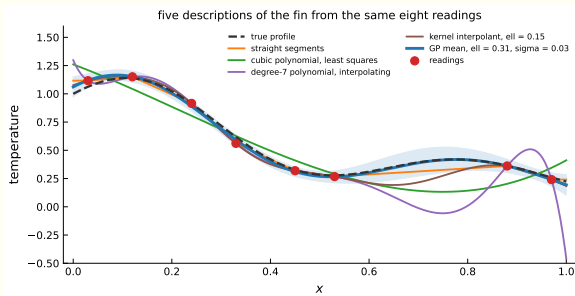
fixed nodes

**a linear solve, with
the projection
theorem's guarantee**

moving nodes

**a generally
non-convex fit,
without the
projection guarantee**

Choosing a representation for the cooling fin



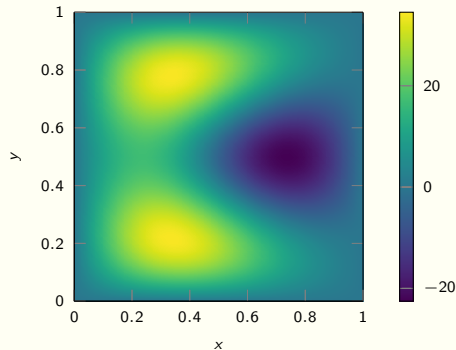
description	relative L^2 (%)	LOO RMSE
straight segments	6.5	
cubic, least squares	23	
degree-7 interpolation	31	
kernel interpolant, $\ell = 0.15$	11	
Gaussian-process mean, $\ell = 0.31$	3.2	
noise standard deviation		

Stored numbers, shape placement, and assumption for each family

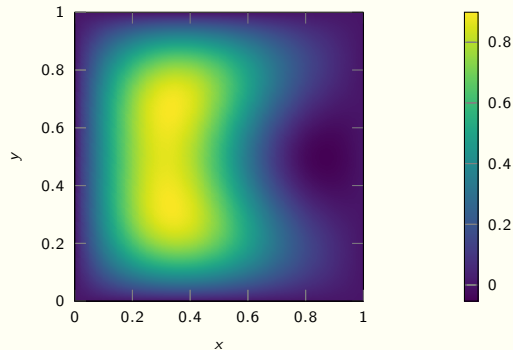
family	stores	shapes placed by	assumes	typical use
stations, grid	values at fixed points	analyst	linear variation between stations	smooth fields, etry
mesh hats	nodal values	analyst and mesher	piecewise-linear variation	complex geom
sine modes	global coefficients	analyst	smooth odd periodic continuation	smooth pinned
B-splines, NURBS	coefficients at fixed knots	analyst and CAD	smoothness across knots	engineered sha
wavelets	selected coefficients and indices	analyst, data selects	few active scales per location	localized featur
POD modes	coefficients on data-derived modes	the snapshot data	a family near a low-dimensional subspace	near-linear fam
sparse dictionary	few atoms and their indices	the optimizer	few active atoms	few active featur
particles, SPH, MPM	positions and field values	the dynamics	material motion determines positions	advection, larg tion
Gaussian splats	centers, covariances, amplitudes	the optimizer	smooth field with concentrated features	concentrated f
level sets	values of an implicit function	analyst and physics	interface as a zero set	moving interfa
kernels, GP	weights, sites, kernel parameters	analyst and data	regularity set by the kernel	scattered data
neural fields	network weights	the optimizer	the network's smoothness bias	later chapters
latent decoders	codes and a decoder	the optimizer	a low-dimensional nonlinear manifold	later chapters

A membrane under a load, $-\Delta u = f$

the load $f(x, y)$



the displacement $u(x, y)$, pinned on the edge

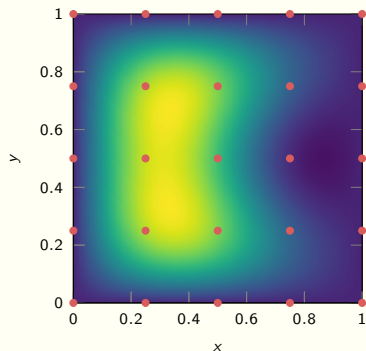


$$-\Delta u = -\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = f \text{ on the unit square,} \quad u = 0 \text{ on its edge}$$

$$u = 0.8 \sin(\pi x) \sin(\pi y) + 0.4 \sin(2\pi x) \sin(\pi y) + 0.2 \sin(\pi x) \sin(3\pi y)$$

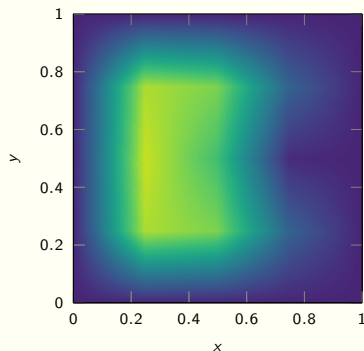
Readings on a grid and bilinear reconstruction

the field u and the 5×5 stations



5×5 , 25 values
largest gap 22.2 percent

bilinear reconstruction from 25 readings

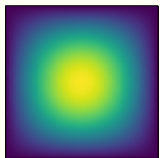


9×9 , 81 values
7.1 percent

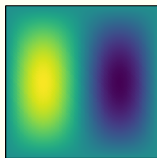
17×17 , 289 values
1.9 percent

Sine modes and hats in two dimensions

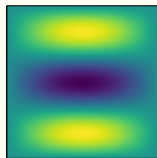
$\sin(\pi x) \sin(\pi y)$



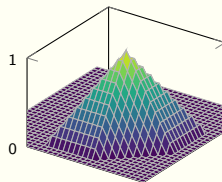
$\sin(2\pi x) \sin(\pi y)$



$\sin(\pi x) \sin(3\pi y)$



a hat on a triangle mesh



$$u(x, y) = \sum_{j,k} a_{jk} \sin(j\pi x) \sin(k\pi y),$$

$$u_h(x, y) = \sum_i u(\mathbf{x}_i) \phi_i(x, y)$$

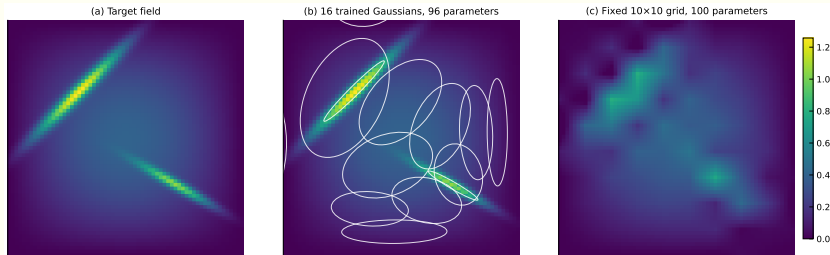
the membrane's weights

$$a_{11} = 0.8, a_{21} = 0.4, \\ a_{13} = 0.2$$

a hat ϕ_i

**one at its node, zero
at the others, linear
on each triangle**

What changes in two and three dimensions



17 stations per direction
17 on a line, 289 on a plate, 4,913 in a solid

a kernel needs only a distance
$$e^{-|\mathbf{x}-\mathbf{x}'|^2/2\ell^2},$$
unchanged

H^1 in two or more dimensions
finite energy, yet possibly unbounded

vector fields
constraints such as
$$\nabla \cdot \mathbf{u} = 0$$

Summary

- 1 Pointwise sampling
- 2 Functions as vectors
- 3 Norms and the meaning of error
- 4 Inner products, projection, and orthogonality
- 5 Changing the basis
- 6 Convergence, density, and completeness
- 7 Kernels and point evaluation
- 8 Noisy observations, ridge regression, and Gaussian conditioning
- 9 Approximating a family of functions
- 10 Adaptive representations
- 11 Choosing a representation for the cooling fin
- 12 From functions to fields